

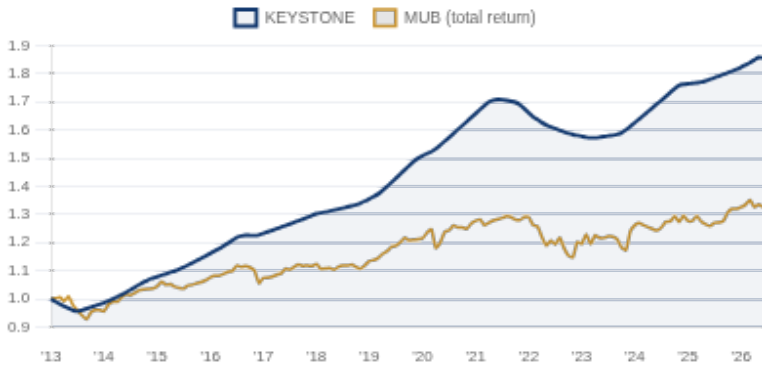
KEYSTONE

Dislocation-reversion in individual municipal bonds — buying forced-seller mispricings on the MSRB regulatory trade tape.

Asset class · US tax-exempt munis (CUSIP-level)
Backtest · 2013–2026
Benchmark · MUB (total return)
Universe · 3,085 bonds · 5.0M prints

+3.2% EXCESS VS CTRL p<0.001	74% WIN RATE 3,089 trades	+4.6% MEAN / TRADE ~1yr, incl coupon	+4.74% CAGR MUB +2.13%	+85.7% TOTAL (13.4Y) MUB +32.5%	-8.0% MAX DD* MUB -13.7%
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GROWTH OF \$1 — TOTAL RETURN VS MUB



	Total	CAGR	Max DD*
KEYSTONE strategy	+85.7%	+4.74%	-8.0%
MUB (total return)	+32.5%	+2.13%	-13.7%

REGIME ROBUSTNESS — EXCESS VS MATCHED CONTROL

Era	Trades	Win	Mean/tr	Excess	p
2013–2016	205	93%	+6.49%	+2.66%	<.001
2017–2019	129	92%	+7.02%	+1.63%	0.00
2020–2021	518	85%	+8.18%	+10.68%	<.001
2022	891	35%	-2.73%	-2.08%	1.00
2023–2025	1375	86%	+6.10%	+4.50%	<.001

THE STRATEGY

Buy when a bond prints **≥3 pts below its own trailing 60-day median** (a forced-seller dislocation) and it clears a trailing-liquidity gate. **Fill** at the next customer-buy print. **Hold ~1 year**; **exit** into a customer-sell print. Coupon accrues throughout.

The dealer bid–ask spread (~0.44 pt round-trip) kills short-term timing but is a one-time cost against growing carry + reversion. Buying idiosyncratic dislocations and holding to reversion beats random entry **in the same bonds** — pure timing alpha, net of spread, carry and selection.

SIGNAL STRENGTH BY DEPTH (FULL SAMPLE)

Entry threshold	Trades	Win	Excess	p
≥1.0pt below median	5,094	70%	+0.38%	<.001
≥2.0pt below median	4,074	72%	+1.57%	<.001
≥3.0pt below median	3,089	74%	+3.25%	<.001

IN-SAMPLE → OUT-OF-SAMPLE (LOCKED)

Window	Trades	Mean/tr	Excess
In-sample 2012–2022	1,701	+2.56%	+1.99%
Out-of-sample 2023–25 (locked)	1,497	+6.40%	+4.76%

KEYSTONE-XL — the validated upgrade

The execution stack proven on corporates transfers verbatim: **limit entry** (fill only if buy price ≤ prior mid + 0.25), **1 position/issuer**, **recovery exit** (first sell print ≥21d once mid recovers to entry-day median). Full window 2012–26: **+7.82% CAGR at -4.8% maxDD** (base +4.35% / -9.1%); OOS 2023–26 **+9.29% CAGR, 98% win**, excess +2.37% (p<0.001) vs a cap-matched control. Validated in four independent samples (muni & corp, IS & OOS each). 2026-08 audit: the issuer cap previously keyed on an id prefix and never bound; it now uses real issuers (book 573 → 336 OOS trades, returns intact). OOS win rates are flattered by right-censored final-year holds and universe survivorship — the durable excess estimate is +2.5–3.6%; live recovery exits should trigger on the prior day's mid (~-0.4 to -0.7pp/trade). See XL_AUDIT.md.

Known failure mode — sustained rate selloffs

Mean-reversion is short a rate shock. In 2022 the whole market repriced down and dislocations did not revert (-1.99% excess). Size accordingly; an optional MUB-trend overlay mitigates it.

DATA & VALIDATION

For institutional discussion only — not investment advice. Simulated results on historical MSRB EMMA public trade data; net of modeled customer bid–ask spread, gross of fees/financing/taxes. *Drawdown is end-of-period peak-to-trough decline in equity value (not annualized); total return is a forward-looking estimate. The outcome of the backtest is not a guarantee of future performance. Source: MSRB EMMA — the regulatory trade tape (every muni trade, tick-level price/yield/size/side); Deep dislocations are rare, ~230/yr in the sampled book; 7,000 bonds trade ≥250×/yr nationally. Daily screen on the fresh EMMA tape flags eligible dislocations. Fills sourced as the patient buyer into forced selling.

CAPACITY & POSITIONING

Patient-capital, capacity-limited sleeve; complements a core muni book.

IMPLEMENTATION

- Customer-buy $\geq$ customer-sell same-day: 96.9%.
- Currently **dormant** — no bond ≥ 3 pt dislocated (calm market).
- Overlays: position sizing, credit screen, MUB-trend regime gate.